

Free Space Green's Functions for the Laplacian

The Poisson Equation

We consider the equation

$$-\Delta u = f(x)$$

in $\mathbb{R}^n$.

Goal: construct a function $G(x)$ such that

$$-\Delta G = \delta(x).$$

Then the solution of the Poisson equation is

$$u(x) = \int_{\mathbb{R}^n} G(x - \xi) f(\xi) d\xi.$$

G is called the **free space Green's function**.

The Laplacian in Divergence Form

The Laplacian can be written as the divergence of the gradient:

$$\Delta u = \nabla \cdot (\nabla u).$$

In coordinates this is

$$\Delta u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2}.$$

This form emphasizes that the Laplacian measures the **net flux of the gradient field**.

Using the divergence theorem,

$$\int_{\Omega} \Delta u \, dx = \int_{\partial\Omega} \nabla u \cdot n \, dS.$$

This identity will be crucial in determining the constants in the Green's functions for the Laplacian.

Symbol of the Laplacian

Take the Fourier transform of the equation

$$-\Delta G = \delta.$$

Using

$$\mathcal{F}(\Delta u) = -|p|^2 \hat{u},$$

we obtain

$$|p|^2 \hat{G}(p) = 1.$$

Thus

$$\hat{G}(p) = \frac{1}{|p|^2}.$$

The function

$$\sigma(p) = |p|^2$$

is the **symbol of the Laplacian**.

Symmetry of the Problem

The equation

$$-\Delta G = \delta$$

is invariant under rotations.

Therefore the Green's function depends only on

$$r = |x|.$$

Thus

$$G(x) = g(r).$$

Rotationally Invariant Functions

The Laplacian

$$\Delta u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2}$$

is invariant under rotations of $\mathbb{R}^n$.

If the source term is rotationally symmetric (for example $\delta(x)$), the Green's function must also be rotationally symmetric.

Thus the solution depends only on the radius

$$r = |x| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

We therefore write

$$u(x) = g(r).$$

Deriving the Radial Laplacian

First derivatives:

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r} \quad \text{and} \quad \frac{\partial u}{\partial x_i} = g'(r) \frac{x_i}{r}.$$

Now differentiate again:

$$\frac{\partial^2 u}{\partial x_i^2} = g''(r) \frac{x_i^2}{r^2} + g'(r) \frac{\partial}{\partial x_i} \left(\frac{x_i}{r} \right).$$

A short calculation gives

$$\frac{\partial}{\partial x_i} \left(\frac{x_i}{r} \right) = \frac{1}{r} - \frac{x_i^2}{r^3}.$$

Summing over i produces

$$\Delta u = g''(r) + \frac{n-1}{r} g'(r).$$

Radial Form of the Laplacian

For a radially symmetric function

$$u(x) = g(r), \quad r = |x|$$

the Laplacian becomes

$$\Delta u = g''(r) + \frac{n-1}{r}g'(r).$$

Thus away from the origin the Green's function satisfies

$$g''(r) + \frac{n-1}{r}g'(r) = 0.$$

Radial Divergence Form of the Laplacian

$$\Delta u = g''(r) + \frac{n-1}{r} g'(r).$$

Multiply by r^{n-1} gives the informative version

$$r^{n-1} \Delta u = r^{n-1} g''(r) + (n-1) r^{n-2} g'(r).$$

The right-hand side is exactly a derivative:

$$r^{n-1} \Delta u = \frac{d}{dr} (r^{n-1} g'(r)).$$

Therefore

$$\Delta u = \frac{1}{r^{n-1}} \frac{d}{dr} (r^{n-1} g'(r))$$

This reflects geometric spreading through spheres of radius r .

Solving the Radial Equation

Multiply by r^{n-1} :

$$r^{n-1}g'' + (n-1)r^{n-2}g' = 0.$$

This gives

$$\frac{d}{dr}(r^{n-1}g'(r)) = 0.$$

Integrating,

$$g'(r) = \frac{C}{r^{n-1}}.$$

General Radial Solutions

Integrating again gives

$$g(r) = \begin{cases} \frac{C}{(2-n)r^{n-2}} + D, & n > 2 \\ C \log r + D, & n = 2 \end{cases}$$

Thus the form of the Green's function depends on the dimension.

Determining the Constant

To determine C we use Gauss' theorem.

Integrate the equation

$$-\Delta G = \delta$$

over a ball B_ε .

$$\int_{B_\varepsilon} -\Delta G \, dx = 1.$$

Applying the divergence theorem gives

$$-\int_{\partial B_\varepsilon} \nabla G \cdot n \, dS = 1.$$

Flux Condition

Since $G = g(r)$,

$$\nabla G \cdot n = g'(r).$$

Thus

$$-\int_{\partial B_\varepsilon} g'(\varepsilon) dS = 1.$$

If ω_n is the surface area of the unit sphere,

$$-\omega_n \varepsilon^{n-1} g'(\varepsilon) = 1.$$

Green's Function in $\mathbb{R}^3$

For $n = 3$

$$g'(r) = \frac{C}{r^2}.$$

The flux condition gives

$$-4\pi C = 1.$$

Thus

$$C = -\frac{1}{4\pi}.$$

Integrating,

$$G(x) = \frac{1}{4\pi|x|}.$$

Green's Function in $\mathbb{R}^2$

For $n = 2$

$$g(r) = C \log r.$$

Using the flux condition

$$-2\pi C = 1.$$

Thus

$$C = -\frac{1}{2\pi}.$$

Therefore

$$G(x) = -\frac{1}{2\pi} \log |x|.$$

Summary

Free-space Green's functions for the Laplacian:

In $\mathbb{R}^3$

$$G(x) = \frac{1}{4\pi|x|}.$$

In $\mathbb{R}^2$

$$G(x) = -\frac{1}{2\pi} \log |x|.$$

Solutions of the Poisson equation are

$$u(x) = \int G(x - \xi) f(\xi) d\xi.$$

The One-Dimensional Green's Function

Consider

$$-\frac{d^2}{dx^2}G(x) = \delta(x)$$

on $\mathbb{R}$.

Away from the origin we have

$$G''(x) = 0.$$

Thus the solution is linear on each side:

$$G(x) = \begin{cases} ax + b, & x > 0 \\ cx + d, & x < 0 \end{cases}$$

Continuity at $x = 0$ gives

$$b = d.$$

Derivative Jump Condition

Integrate the equation across the origin:

$$\int_{-\varepsilon}^{\varepsilon} -G''(x) dx = \int_{-\varepsilon}^{\varepsilon} \delta(x) dx = 1.$$

The left side becomes

$$-(G'(\varepsilon) - G'(-\varepsilon)) = 1.$$

Thus

$$G'(0^+) - G'(0^-) = -1.$$

Since $G'(0^+) = a$ and $G'(0^-) = c$,

$$a - c = -1.$$

Choosing the symmetric solution gives

$$G(x) = -\frac{1}{2}|x|.$$

Summary of Free-Space Green's Functions

For the Laplacian in $\mathbb{R}^n$:

$$-\Delta G = \delta$$

$$G(x) = \begin{cases} -\frac{1}{2}|x|, & n = 1 \\ -\frac{1}{2\pi} \log |x|, & n = 2 \\ \frac{1}{4\pi|x|}, & n = 3 \end{cases}$$

Each case reflects how flux spreads through spheres:

- ▶ $n = 1$: two directions
- ▶ $n = 2$: circumference grows like r
- ▶ $n = 3$: surface area grows like r^2